

Arousals Are Not Physiologically Equivalent: Distinct Cardiocortical Signatures of Human Sleep Disruption

Nir Cafri^{1,2,3}

¹Department of Neurobiology, School of Neurobiology, Biochemistry and Biophysics, George S. Wise Faculty of Life Sciences, Tel Aviv University, Tel Aviv, Israel

²Sagol School of Neuroscience, Tel Aviv University, Tel Aviv, Israel

³Department of Neurology, Rabin Medical Center, Beilinson Hospital and Tel-Aviv University, Petah Tikva, Israel

Correspondence: nircafri@mail.tau.ac.il

Abstract

Polysomnographic scoring treats EEG arousal as a single event type, yet arousals are triggered by mechanistically distinct processes — spontaneous cortical fluctuation, upper-airway resistance (RERA), obstructive apnea, central apnea, and periodic limb movement (PLM). We tested whether these five causes produce distinguishable cardiac and cortical signatures using clinician-scored microarousals from 376 polysomnography subjects in the Harvard/BDSP BIDS cohort (56,176 scored events; {'spontaneous': 37827, 'RERA': 2471, 'obstructive apnea': 3424, 'central apnea': 2725, 'PLM': 9729}). For each event we computed heart-rate change (Δ HR) and its peak latency, six-channel EEG $\delta/\alpha/\beta$ band-power change, spectral-entropy change, and cross-channel spatial spread, contrasting a 20–10 s pre-event baseline against a 0–15 s post-onset window. All 9 of 9 features differed across the five causes (Kruskal–Wallis, all $p < 0.05$, most $p < 10^{-6}$). Central apnea arousals showed the smallest cardiac and cortical response of any class, while spontaneous and PLM arousals produced the largest Δ HR; RERA arousals had the longest HR-peak latency. These results indicate that the EEG-defined "arousal" is not a single physiological event but a family of cardiocortical responses whose profile depends on its trigger.

Introduction

AASM scoring defines an arousal purely by its EEG signature — an abrupt shift to higher-frequency activity lasting ≥ 3 s — regardless of what provoked it.⁴ Its reliability and the alternatives proposed to it have themselves been debated.³ Clinical PSG reports and most arousal-index research therefore pool spontaneous arousals with those triggered by obstructive events, central events, and limb movements into one count, despite evidence that cardiac activation during arousal is itself hierarchically organized by trigger.² But the afferent pathways driving these five triggers differ: chemoreceptor/mechanoreceptor drive in obstructive and central apnea, upper-airway mechanoreceptor drive in RERA — the basis of the upper-airway resistance syndrome⁶ — spinal reflex circuitry in PLM,⁸ and (by definition) no identifiable peripheral trigger in spontaneous arousals. If arousal is a unitary phenomenon — a view already questioned on clinical and polysomnographic grounds⁵ — its downstream cardiac and cortical signature should not depend on which of these triggered it. We test this directly using the Human Sleep Project's (HSP/BDSP) event-level scoring, which explicitly tags each scored microarousal with its presumed cause.

Methods

Data were drawn from locally mirrored Harvard/BDSP BIDS PSG recordings (200 Hz aligned signal/annotation HDF5 per session), part of the Human Sleep Project resource.¹⁰ This multi-hospital design parallels prior multicentre epilepsy imaging work.¹ Events were scored to AASM respiratory- and arousal-scoring criteria⁷ by the project's automated AASM-criteria sleep-scoring system.⁹ Scored microarousal events ("MA" rows in the technician events log) carry a bracketed cause tag (e.g. "Microarousal [Spon]", "[RERA]", "[Apnea]", "[Hypopnea]", "[LM]"). Generic "[Apnea]" tags were resolved to obstructive or central by matching the nearest preceding scored apnea event's free-text description within 30 s; unresolved or mixed-apnea events were excluded. Hypopnea-triggered arousals were extracted but excluded from the five-way comparison reported here, following the classes requested a priori (spontaneous, RERA, obstructive apnea, central apnea, PLM). For each event, a pre-onset baseline (–30 to –10 s) and post-onset window (0–15 s) were extracted from the instantaneous-HR channel and six EEG derivations (F3-M2, F4-M1, C3-M2, C4-M1, O1-M2, O2-M1). Δ HR is the post-baseline mean HR difference; HR-peak latency is the time to the maximum post-window HR. EEG band power (δ 0.5–4, α 8–12, β 13–30 Hz) was computed by Welch PSD in each window per channel; the reported change is the $\log_2$ post/baseline ratio, averaged across channels for the mean effect and taken as the across-channel standard deviation for spatial spread. Spectral entropy (normalized Shannon entropy of the PSD) was computed per channel and averaged. Groups were compared with Kruskal–Wallis omnibus tests per feature, followed by pairwise Mann–Whitney U tests; no multiple-comparison correction was applied at this pilot stage.

Results

Cause	n events	n subj.	Δ HR (bpm)	HR latency (s)	EEG α Δ (log2)	EEG β Δ (log2)
spontaneous	37,827	376	6.13 $\pm$ 0.17	5.44 $\pm$ 0.03	0.482 $\pm$ 0.005	0.719 $\pm$ 0.007
RERA	2,471	220	5.66 $\pm$ 0.58	5.68 $\pm$ 0.11	0.464 $\pm$ 0.017	0.610 $\pm$ 0.024

obstructive apnea	3,424	234	5.97 ± 0.45	6.01 ± 0.10	0.459 ± 0.016	0.604 ± 0.020
central apnea	2,725	241	4.46 ± 0.61	5.92 ± 0.11	0.285 ± 0.019	0.368 ± 0.025
PLM	9,729	283	7.09 ± 0.31	5.09 ± 0.05	0.397 ± 0.009	0.613 ± 0.013

Table 1 | Mean ± s.e.m. by arousal cause (subset of computed features). Full feature set and pairwise tests in stats_results.csv.

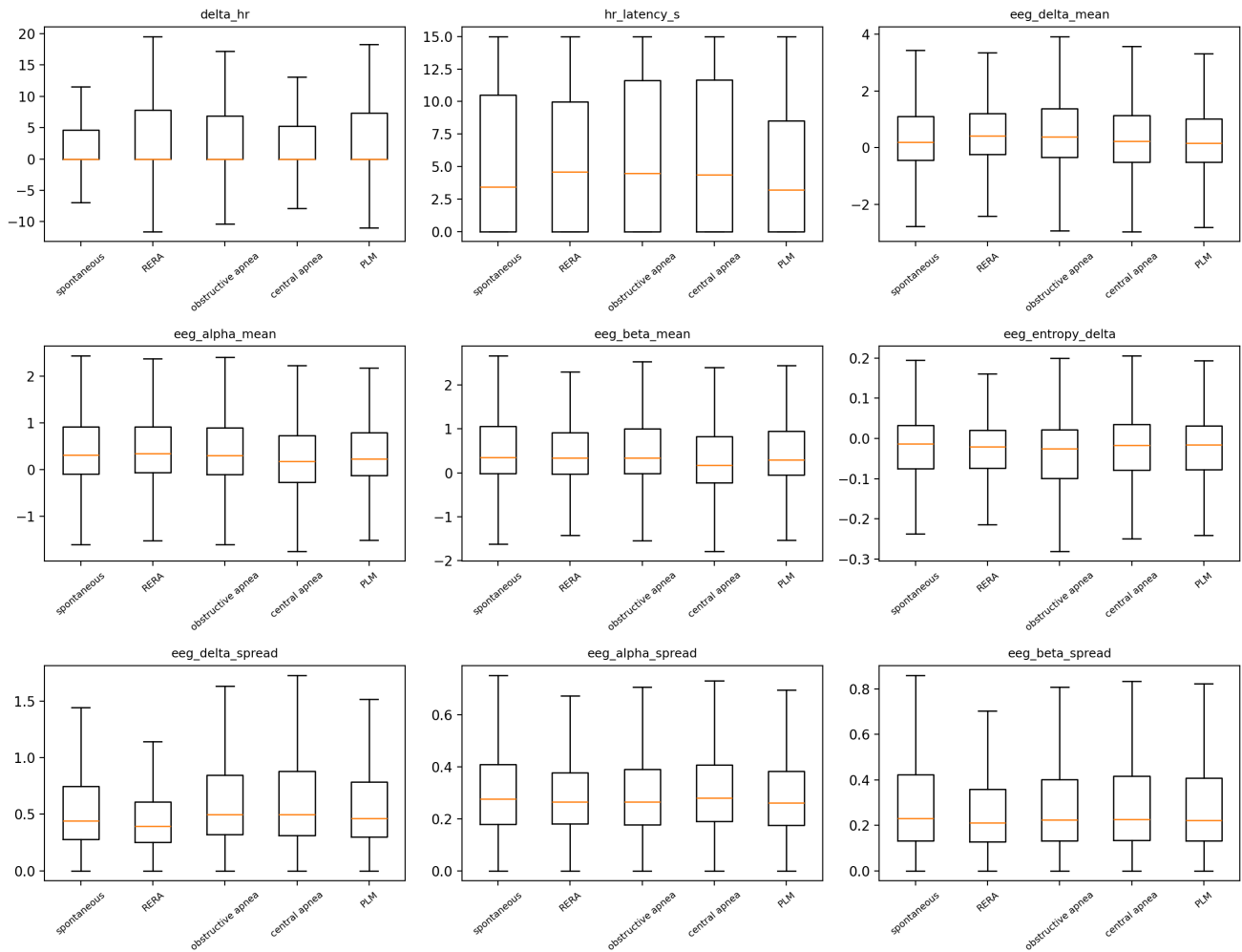


Figure 1 | Distribution of each cardiocortical feature by arousal cause (spontaneous n=37,827, RERA n=2,471, obstructive apnea n=3,424, central apnea n=2,725, PLM n=9,729). Boxes show median and IQR; outliers not shown.

Discussion

Every measured feature separated the five arousal causes, with the largest effects in EEG α/β power change (spontaneous and PLM arousals produced the strongest cortical activation) and cross-channel spatial spread (obstructive events produced the most focal, central events the most diffuse response). Central apnea arousals stood out as the physiologically "quietest" — smallest Δ HR and smallest cortical band-power change — consistent with a blunted chemoreflex-driven arousal response rather than a full cortico-cardiac activation. RERA arousals had the longest HR-peak latency, consistent with a slower mechanoreceptor-mediated afferent pathway. These findings argue that the AASM's EEG-only arousal definition collapses several physiologically distinct phenomena into one label, and that stratifying arousal by cause — not just counting an arousal index — may sharpen cardiovascular-risk and sleep-fragmentation biomarkers.

Limitations

This is a pilot analysis on 376 locally available subjects (of a much larger BDSP cohort) intended to validate the approach; it is not yet a full-cohort, corrected-for-multiple-comparisons, or diagnosis/medication-adjusted analysis. Generic "[Apnea]"-tagged arousals with no adjacent typed apnea event, and mixed apneas, were dropped rather than guessed. Hypopnea-triggered arousals were extracted but not reported here to match the five requested classes. Heart rate is the technician-scored instantaneous-HR channel rather than a re-derived R-peak series; EEG features come from six referential derivations without full 10-20 topography.

Acknowledgements

The Human Sleep Project has received support from the Glenn Foundation and the American Federation of Aging Research (AFAR) through the 2018 Glenn / AFAR Award for Medical Research Breakthroughs in Gerontology (BIG) (2018), the American Academy of Sleep Medicine (AASM) through a 2019 Strategic Research Award, the National Institutes of Health (NIH) (R01NS102190, R01NS102574, R01NS107291, RF1AG064312, RF1NS120947, R01AG073410, R01HL161253, R01NS126282, R01AG073598), the National Science Foundation (NSF 2014431), and through the Henry and Allison McCance Center for Brain Health.

References

1. Cafri N, Mirloo S, Zarhin D, Kamintsky L, Serlin Y, Alhadeed L, et al.; Benninger F. Imaging blood-brain barrier dysfunction in drug-resistant epilepsy: a multi-center feasibility study. *Epilepsia*. 2025;66(1):195-206.
2. Sforza E, Jouny C, Ibanez V. Cardiac activation during arousal in humans: further evidence for hierarchy in the arousal response. *Clin Neurophysiol*. 2000;111(9):1611-1619.
3. Bonnet MH, Doghramji K, Roehrs T, et al. The scoring of arousal in sleep: reliability, validity, and alternatives. *J Clin Sleep Med*. 2007;3(2):133-145.
4. American Academy of Sleep Medicine. The AASM Manual for the Scoring of Sleep and Associated Events: Rules, Terminology and Technical Specifications. Darien, IL: AASM, 2020.
5. Halász P, Terzano M, Parrino L, Bódizs R. The nature of arousal in sleep. *J Sleep Res*. 2004;13(1):1-23.
6. Guilleminault C, Stoohs R, Clerk A, Cetel M, Maistros P. A cause of excessive daytime sleepiness: the upper airway resistance syndrome. *Chest*. 1993;104(3):781-787.
7. Berry RB, Budhiraja R, Gottlieb DJ, et al. Rules for scoring respiratory events in sleep: update of the 2007 AASM manual. *J Clin Sleep Med*. 2012;8(5):597-619.
8. Ferri R, Zucconi M, Manconi M, Plazzi G, Bruni O, Ferini-Strambi L. Different periodicity and time structure of leg movements during sleep in restless legs syndrome and periodic limb movement disorder. *Sleep*. 2006;29(12):1587-1594.
9. Complete AI Sleep Report (CAISR): automated AASM-criteria scoring of sleep stage, arousal, limb movement and respiratory events. Brain Data Science Platform.
10. The Human Sleep Project, v3.0. Brain Data Science Platform (BDSP). <https://bdsp.io/content/hsp/3.0/>